

Uses of REST APIs 30m

The three common uses of REST APIs are:

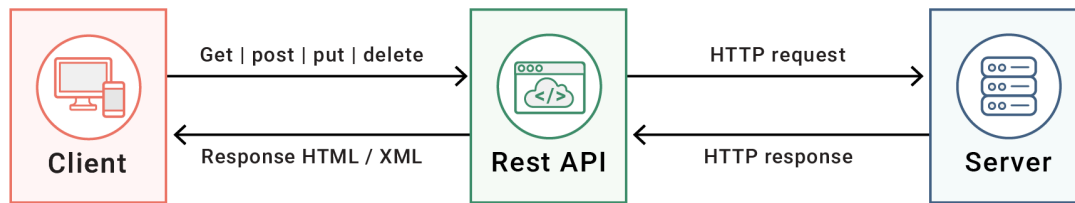
- cloud applications
- cloud computing services
- web uses.

For each application and use, it is necessary to establish a connection between the client and server sides.



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The API connection between the client and server side



The following architectural requirements need to be satisfied to consider an API a REST API:

- **A client-server architecture:** The client-server architecture has resources and manages requests. The server manages application data and state and communicates with a client to handle the user's requests. There is a clear separation that must be present.
- **Stateless:** There should not be any client information stored between requests, and every request is independent in terms of connection.
- **Cacheable data:** Servers must have marked their responses as cacheable or not so that the client can cache them whenever possible to improve performance.
- **Uniform interface:** Services should expose data as resources using a consistent naming convention, including standardized URLs and HTTP methods (e.g., GET, POST). Each request should include enough information to describe how it should be processed, and responses must use appropriate HTTP status codes to indicate the outcome.
- **Layered system:** Invisible to the client, a system in which each type of server is organised into hierarchies.

The code below illustrates this type of client-server communication. A user wants to know their geographical location, so they get the location from the server using a Geolocation API. This is an API within a browser that provides location data to web applications. It does require user permissions when used.

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```
1  <!DOCTYPE html>
2  <html>
3  <body>
4
5  <h2>JavaScript Geolocation API</h2>
6  <p>Click the button to get your coordinates.</p>
7
8  <button onclick= "getLocation() ">Try It</button>
9
10 <p id= "demo "></p>
11
12 <script>
13 const x = document.getElementById( "demo ");
14
15 function getLocation() {
16 if (navigator.geolocation) {
17   navigator.geolocation.watchPosition(showPosition);
18 } else {
19   x.innerHTML = "Geolocation is not supported by this browser. ";
20 }
21 }
22
23 function showPosition(position) {
24   x.innerHTML= "Latitude: " + position.coords.latitude +
25   "<br>Longitude: " + position.coords.longitude;
26 }
27 </script>
28
29 </body>
30 </html>
```



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API

Non-Assessed

Copy and execute the above code on your local server using either a XAMPP or WAMP server, and then answer these questions:

1. Identify the type of request placed by the client to the server.
2. Why does this API require user permissions.

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Save draft

Submit

☒ REST APIs

Non-Assessed

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There are many applications that use REST APIs. Some you may use on a day-to-day basis without even realising that they use REST APIs. Identify two applications that use REST APIs.

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0 / 250 Word Limit

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Thoughts on this page?



REST APIs

Can you share two apps you use every day that secretly rely on REST APIs.

B <i>I</i>
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Jorge Andres Guerrero Giraldo 22 Feb 2026, 11:40 AM

I use **Instagram** and **Google Maps** every day, and both rely on REST APIs. Instagram fetches feeds, profiles, comments, and likes through HTTP requests that return JSON. Google Maps requests map tiles, places, and directions from web services so it can display locations and routes in the app.

